

Lesson 4.4 Put Our Plan into Action

Location: Classroom

Grade Level: High school

Time: Two class sessions of 50 minutes

Driving Question

- How can we put our ideas into action to create a science communication product?

Lesson Synopsis

In this lesson, students will refine their science communication products based on the feedback they receive from Module 4.3. They will build the products (e.g., with Canva) to convey their climate issue, their key message, and supporting evidence to the chosen audience. By the end of the lesson, students should have a more polished draft of their products that can be presented to the target audience.

What students will figure out:

- Science communication products are like models. They simplify complex systems, highlight important features, and are built for a purpose to influence or inform a specific audience.
- Refining a product based on feedback makes communication more effective. What wasn't clear before can be adjusted to better reach the target audience.
- Communicating climate science requires linking human activities to climate impacts in ways that connect to people's real-world concerns and choices, in order to inform action.

LEARNING OUTCOMES

By the end of this investigation, students will be able to...	You can assess this using...
1. Articulate how they will revise their product, based on the feedback they received.	Group discussion in the Navigate section.
2. Develop a science communication product that clearly conveys a climate issue, a key message, and includes supporting evidence.	Science communication product in the Investigate section.

Building towards NGSS and AI Literacy:

- **HS-PS2-6:** Communicate scientific and technical information (e.g. about the process of development and the design and performance of a proposed process or system) in multiple formats (including orally, graphically, textually, and mathematically).

LESSON OVERVIEW

Section	Time	What Happens
Navigate	10 minutes	Students gather feedback from Module 4.3 and articulate what they want to change about their science communication plan.
Investigate	30 minutes	Students create their science communication product.
Share	10 minutes	Students conduct a virtual gallery walk to provide feedback on their peers' projects.
Navigate (OPTIONAL)	5 minutes	Students reflect on how they want to improve their science communication plan.

MATERIALS

- [Lesson 4.4 slides](#)
- [Science journal](#)
- A shared Google Folder where students can upload copies (pdf, images) of their science communication products.
- Canva: <https://www.canva.com/>
 - Infographic templates: <https://www.canva.com/infographics/templates/>
 - Instagram templates: <https://www.canva.com/instagram-posts/templates/>
 - Board game idea templates: <https://www.canva.com/templates/?query=board-game-interact>
- If students are creating a board game, they could use GenAI (e.g., ChatGPT) to help them refine the rules of the game.
- [Teacher Rubric](#) for science communication products (referenced throughout module 4)
 - [Student Rubric Checklist](#) (provided in science journal, p.60)
- [AISciComm chatbot platform](#) (to test messages from Module 4.3)

BEFORE YOU START TEACHING

1. Review the materials, including the Science Communication Product [Student Checklist](#) (also in Science Journal p. 60) and [Teacher Rubric](#) to know the expectations of students.

2. Review possible student template options (see Materials).
3. Throughout the program, students will be working in teams of three (preferably).
4. Make sure that you have a way to project a video with sound.
5. The science journal is intended to be a digital notebook. However, if you decide to print copies for students, they may need a writing instrument (pen, pencil, etc.).

WHERE WE ARE GOING AND NOT GOING

Where We Are Going: Students will refine their science communication products using feedback from Module 4.3. They will build the actual products (e.g., Canva infographics or social media posts, board game rules) to communicate their message. They will practice how to turn climate data and models into compelling, audience-friendly communication. The lesson sequence illustrates to students how **science communication is an iterative design process** that involves cycles of building, testing, and revising.

Where We Are NOT Going: We are not simply making something “pretty”. The purpose is effective communication grounded in evidence, not just aesthetics. *This is the last lesson of the AISciComm curriculum.* We will not be testing with real audiences, but students will have an opportunity to showcase their products with peers in a Gallery Walk activity.

LEARNING SEQUENCE

Lesson plan notes are also provided on the slide notes.

NAVIGATE: Refine your Science Communication Message (10 minutes)

1. Open the slideshow to **slide 1** and introduce the lesson. Explain to students that now that we have tested their message with a chatbot, they will reflect on the interactions and refine their science communication plan.
2. In their project team, students refine their science communication plans based on the conversations they had with the target audience chatbots in Module 4.3. Advance to **slide 2** to show students the guiding questions for **Refine your Message**:
 - *Based on the conversations with the target chatbot, what will you change about your science communication plan? Why?*
 - *What seems to work well with this audience? What evidence do you have from the conversations that supports this claim?*
3. Invite student teams to share their responses. **Possible student ideas to listen for**:
 - We realized our audience didn’t understand some scientific terms, so we need to simplify our language.

- The chatbot responded well to visuals, so we will include more graphs and images.
 - Our message felt too negative; we want to add hopeful solutions to keep the audience engaged.
 - Our slogan worked really well. The chatbot repeated it back, which shows it was memorable.
4. Then, advance to **slide 3** and invite students to discuss as a class: **Testing your Message with a Broader Audience:**
- You were building the chatbot based on one target audience group. *How can we create opportunities for other audience groups to provide feedback on the science communication plan?*

Possible student ideas to listen for:

- We could test our message with classmates who aren't in our group to see if it makes sense to them.
 - We could show our infographic to family members or younger students for feedback.
 - We could run a quick survey (online or paper) with peers.
 - We could recruit other teachers.
5. Advance to **slide 4** and invite students to share ideas as a class:
- Define Success:**
- *How will you know if you are successful with your science communication plan?*
 - *How will you be able to tell if your audience has learned or begun to feel the way that you want them to?*

Possible student ideas to listen for:

- If the audience can repeat back our key message in their own words, we know it landed.
 - If they show interest or ask follow-up questions, that's a good sign.
 - If they can identify at least one action they can take, that shows impact.
6. Advance to **slide 5** and reference the student checklist for science communication products.
- **Slide 6:** Highlight for students that different science communication products may have different success criteria. Engagement for an online Instagram post that is measured by comments will be easier to gauge than a paper infographic.

INVESTIGATE: Build our Science Communication Product (30 minutes)

1. Advance to **slide 7**. In small groups (from previous lessons), students create their science communication products based on the plans that they have created. Students can draft

the products in [Canva](#). [Science Journal p. 72](#) shares directions and additional information.

PUTTING PIECES TOGETHER: Virtual Gallery Walk (10 minutes)

1. As students finish up their products, let them know that their next step will be to conduct a **virtual gallery walk** to see what others have worked on.
2. Advance to **slide 8**. Give students instructions to download a copy of their science communication products and upload them to a shared Google Folder.
3. Assign each project team to review the science communication product of another team. Have students write down on post-it notes or PearDeck their reviews:
 - a. Which project team did you review?
 - b. What is the key message that the product seems to convey?
 - c. What part of this product helped you understand how climate change is impacting the Southern California ocean?
 - d. What did this group do well?
 - e. What suggestions do you have to make this product more complete, engaging, or clear?

NAVIGATE: Reflect on our plans (5 minutes - OPTIONAL)

1. Advance to **slide 9**. As students finish up their gallery walk, invite them to look back at their product. Ask student teams to answer the following questions as a group and then fill out their [Science Journal \(p.73\)](#).
 - What patterns did you notice as you conducted your gallery walk?
 - Were there any ideas or aspects you saw in other products that you would like to incorporate into yours?
 - What surprised you about the ways teams communicated the same climate issue?